

Sagar Modi

Pre-Final Year Undergraduate | B.Tech Computer Science & Design
I.K. Gujral Punjab Technical University, Jalandhar

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EDUCATION

B.Tech – Computer Science & Design 2023 – 2027
SVIET, Chandigarh
CGPA: **8.5 / 10**

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Data Analytics & Visualization

Intermediate (XII) 2020 – 2022
Jharkhand Academic Council
Percentage: **86.80%**

Matriculation (X) 2018 – 2020
Jharkhand Academic Council
Percentage: **86.80%**

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, C++
- **Generative AI & NLP:** Prompt Engineering, Chatbot Development, LLM Applications, Understanding of NLP Concepts, Tokenization, Embeddings, Transformers
- **Data Science & Analytics:** Pandas, NumPy, Scikit-learn, EDA, Feature Engineering, Statistical Analysis, Data Cleaning, Data Visualization
- **Data Visualization:** Excel, Power BI, Matplotlib, Seaborn, Dashboards, Reporting
- **Backend & APIs:** FastAPI, Flask, REST APIs
- **Databases:** MySQL, Supabase, MongoDB, PostgreSQL
- **Additional Skills:** MERN Stack, JavaScript, C, C++
- **Tools:** Git, GitHub, Jupyter Notebook

ACHIEVEMENTS & TRAINING

- **35th Rank** – CodeCrafters Challenge 2025
- **Research Intern** – IIT Roorkee
- **NASA ARSET Trainee** – Satellite Laser Altimetry (ICESat-2)
- **IIRS-ISRO Trainee** – Remote Sensing & Image Analysis
- **SSB Interview Candidate (TES Entry – Indian Navy)** – Selected for officer-level assessment at SSB Kolkata

PROFESSIONAL EXPERIENCE

Research Intern – Software Developer Dec 2025 – Jan 2026
Indian Institute of Technology (IIT) Roorkee

- Worked under the guidance of **Dr. Basant Yadav**, Associate Professor, IIT Roorkee.
- Contributed to development of **JalAI**, an AI-powered water intelligence platform for real-time monitoring and analytics.

- Managed backend infrastructure using **Supabase (PostgreSQL)** and developed real-time data pipelines.
- Processed and analyzed IoT sensor datasets using **Python**, Pandas, and NumPy for **EDA** and anomaly detection.
- Built APIs, **analytics dashboards**, and AI-driven alerting workflows for monitoring systems.
- Rapidly adapted to Flutter-based development despite prior MERN stack background, demonstrating strong learning and problem-solving ability.

🏆 *Research Internship Completion Certificate Awarded*

Project Evaluator & Mentor Oct 2025 – Present
The Uniques Community

Frontend Web Developer Intern Jul 2024 – Aug 2024
CodeAlpha Tech Solutions

PROJECTS

JalAI – AI-Powered Water Intelligence Platform

jalaitech.com

- Developed an **AI-powered** water monitoring platform using real-time **IoT sensor data**.
- Managed backend infrastructure using **Supabase (PostgreSQL)** and built live **data pipelines**.
- Processed sensor datasets using **Python, Pandas, NumPy**, and performed **EDA** for anomaly detection.
- Built **analytics dashboards**, alerting systems, and contributed to **SEO optimization**.
- **Tech Stack:** Python, Pandas, NumPy, PostgreSQL, Supabase, Flutter

The Uniques Community Platform

theuniques.in

- Developed a scalable student community platform with **AI-powered chatbot integration**.
- Built **analytics dashboards** to visualize student **SGPA trends** and skill distribution insights.
- Developed backend **APIs** and **data-driven modules** for student engagement tracking and personalized experiences.
- **Tech Stack:** MERN Stack, MongoDB, REST APIs

InnoSign – AI-Based Sign Language Interpretation System

InnoSign GitHub

- Developed a real-time **sign language interpretation system** using **OpenCV** and **MediaPipe**.
- Implemented **gesture recognition** and developed a **Flask backend** for real-time prediction.
- Improved prediction stability using **temporal smoothing techniques**.
- **Tech Stack:** Python, Flask, OpenCV, MediaPipe, NumPy